

TrES-3b

R-2196

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_121002 · created 2026-08-14T00:16:11+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456202.6645 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.193 +/- 0.047
Transit depth (Rp/Rs)^2	3.71 +/- 1.79 [%]
Orbital Inclination (inc)	81.05 +/- 2.38
Airmass coefficient 1 (a1)	1.207 +/- 0.02
Airmass coefficient 2 (a2)	-0.1367 +/- 0.0099
Scatter in the residuals of the lightcurve fit is	1.69 %
Best Comparison Star	None
Optimal Aperture	3.5
Optimal Annulus	6.17
Transit Duration (day)	0.043 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.193 ± 0.047 vs 0.16309 (deviation 0.54σ)
Duration fit vs published	0.043 d vs 0.0567 d (deviation 0.43σ)
Mid-time O–C	1.8 min
Depth SNR	3.5
Residual scatter	1.69 %

Light curve

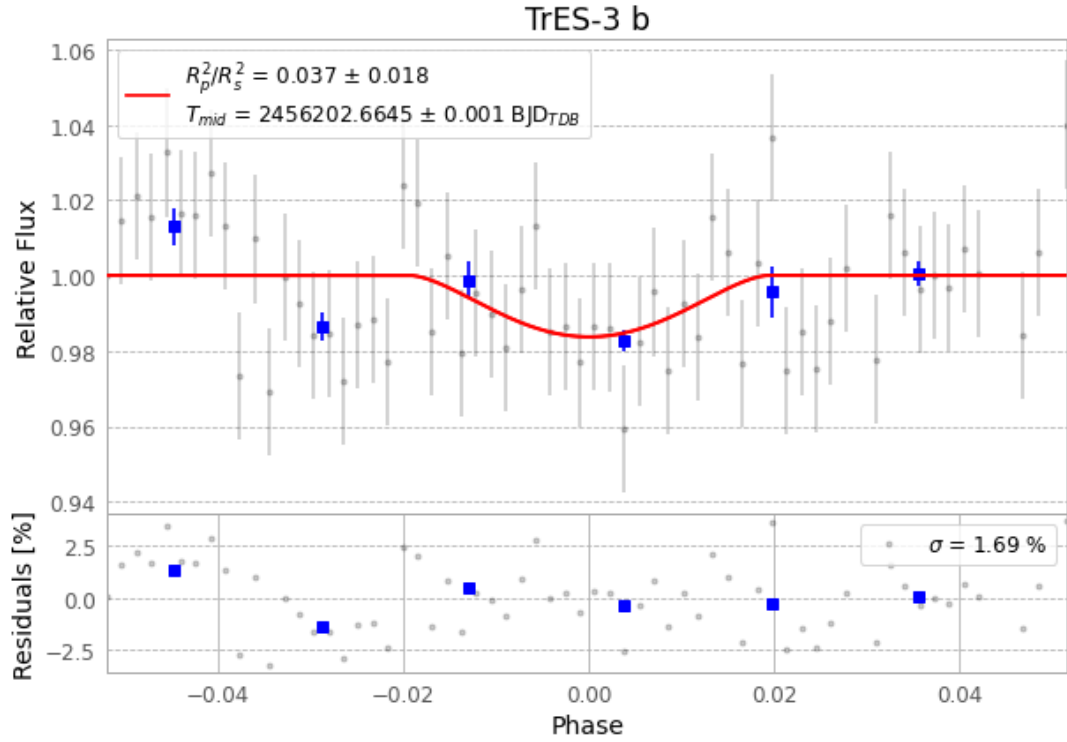


Figure 1 — FinalLightCurve_TrES-3 b_2012-10-01.png (SHA-256 ff58f52f1e70e5c3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [271, 273], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c12de55cd5cbad88...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

64 FITS frames (42 MB), TRES-3121002021812.FITS → TRES-3121002053312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-121002.log (b74bd3cd000d...), FinalLightCurve_TrES-3 b_2012-10-01.png (ff58f52f1e70...), FinalLightCurve_TrES-3 b_2012-10-01.csv (7c863a307f56...)

Compute resources

Wall time 123.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-14 00:16Z.